# OpenClaw Servo Control Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only covers servo-related content.

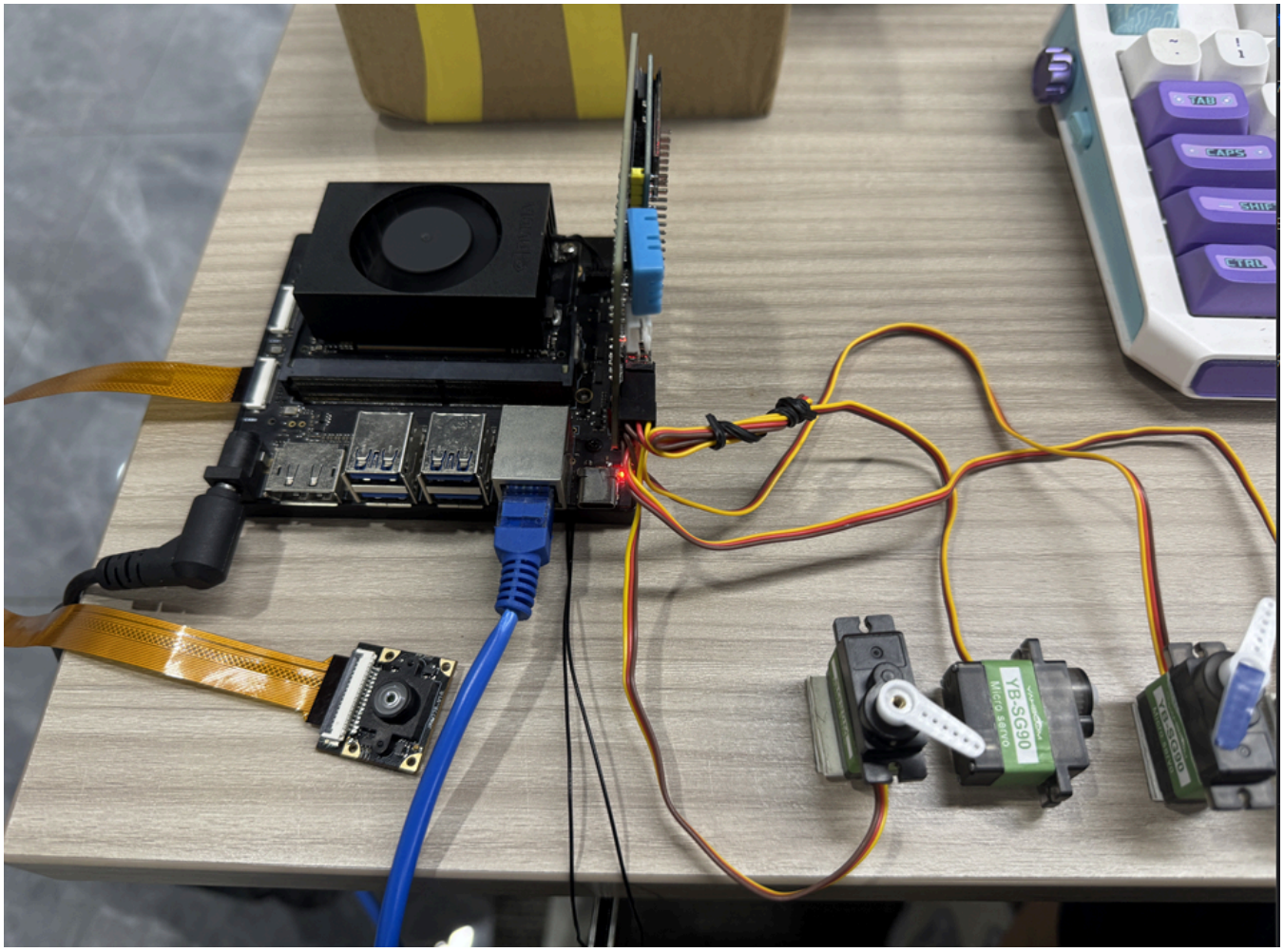
## 1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capability through servo peripherals. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw controlling servos
- Chat with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four servo examples: single servo control, servo dancing, yes/no Q&A, simulated table tennis, simulated pointer

## 2. Hardware Preparation

- OpenClaw main control board
- Servo `[1 / 4 units]`
- `[Optional] Microphone module`
- Wiring diagram




# 3. Software Preparation

**Code entry points used in this tutorial**

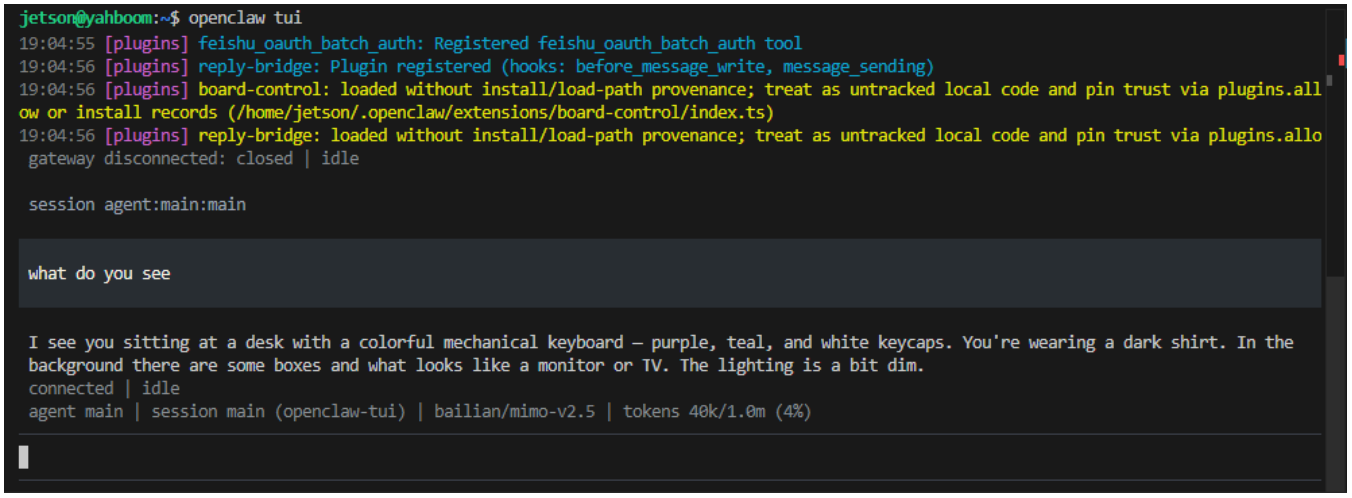
- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- Python driver library: `/home/jetson/Downloads/python/`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

# 4. Testing the Servo

Before starting this tutorial, please complete a basic `openclaw tui` chat self-check; if you haven't done so, see [06-Three Dialogue Configuration Tests] first. After confirming OpenClaw can reply normally, proceed to the servo-specific test.

Enter the following command in the terminal to enter TUI:

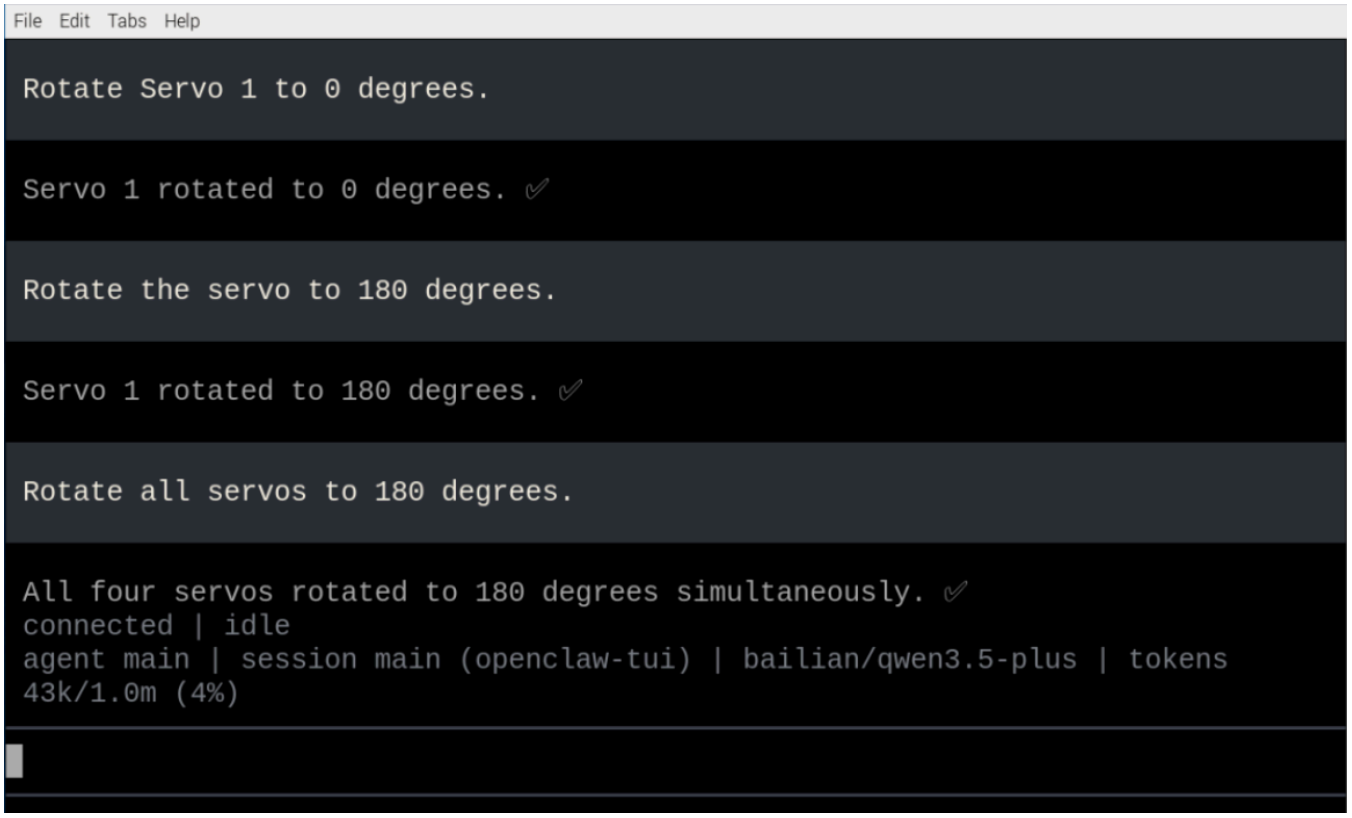
```
openclaw tui
```




After ensuring the hardware is connected, you can test servo control by entering the following in the terminal:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Turn the servo to 180 degrees

Note: If possible, specify which servo number to control




If you can see the servo driving normally, it means all of the following are working properly:

- OpenClaw can receive commands
- Servo power supply is normal
- Servo angle control range is correct
- Clear feedback is visible after command execution

# 5. Three Methods to Chat with OpenClaw

| Method | Suitable Scenarios | Advantages | Recommended Positioning |
| --- | --- | --- | --- |
| WhatsApp | Remote control, classroom demo | Easy to connect, suitable for multi-person experience | Demonstrate "remote natural language control" |
| TUI | Local debugging, command verification | Most direct, easiest to troubleshoot | First get the control pipeline working |
| Voice Code | Voice interaction, complete AI experience | Closest to real conversation experience | As the final demonstration method |

**If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.**

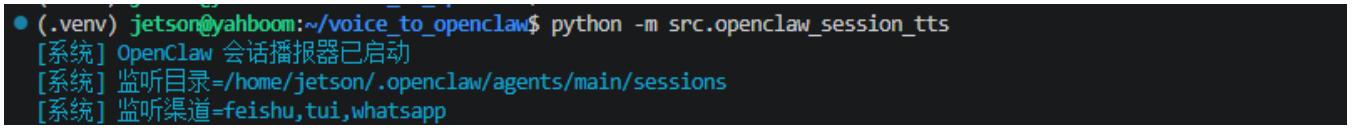
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

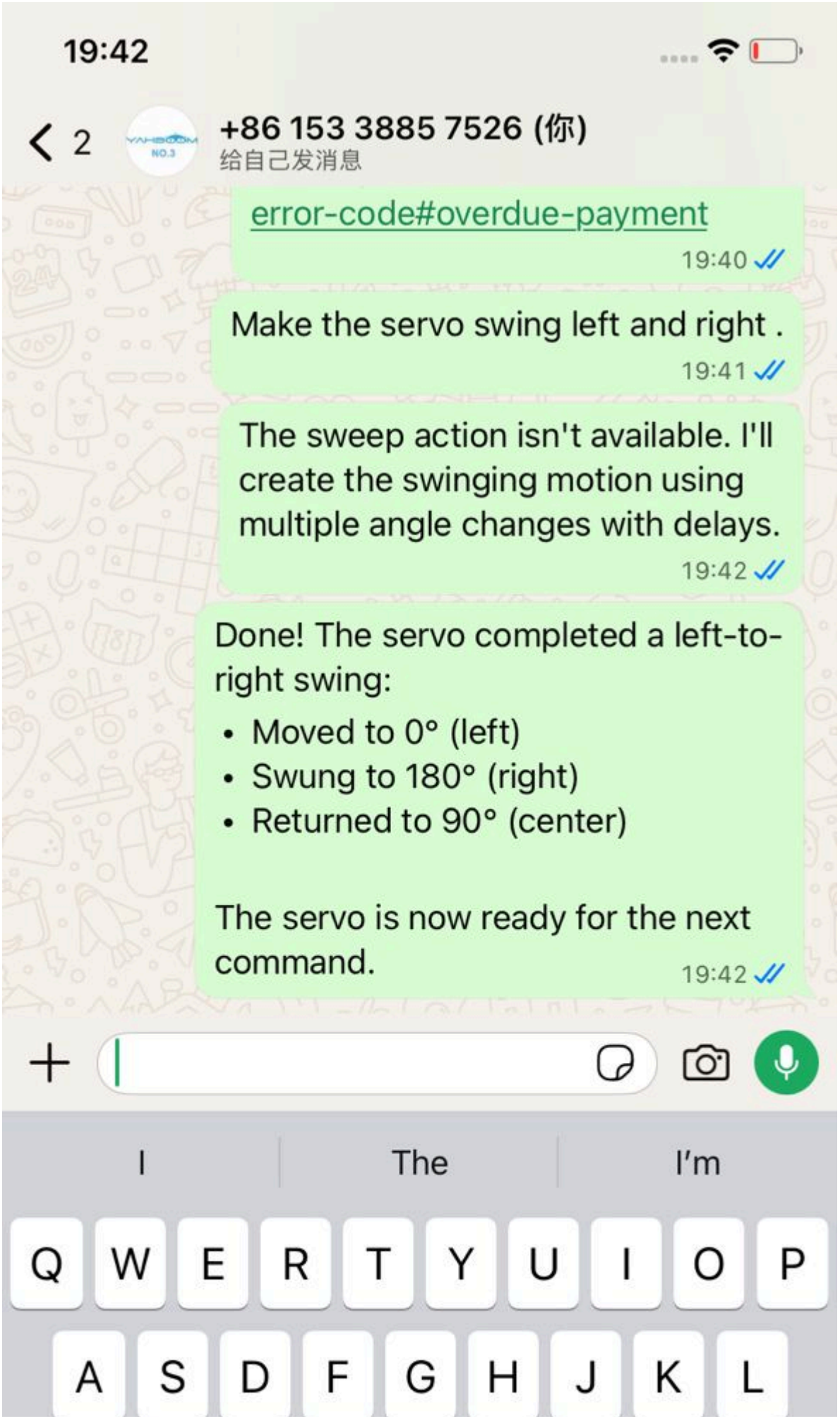



## 5.1 whatsapp Method

For WhatsApp connection, QR code creation, and gateway restart steps, please refer to [06-Three Dialogue Configuration Tests]. This section only retains chat examples suitable for the servo tutorial.

You can chat directly, for example:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Make the servo swing left and right

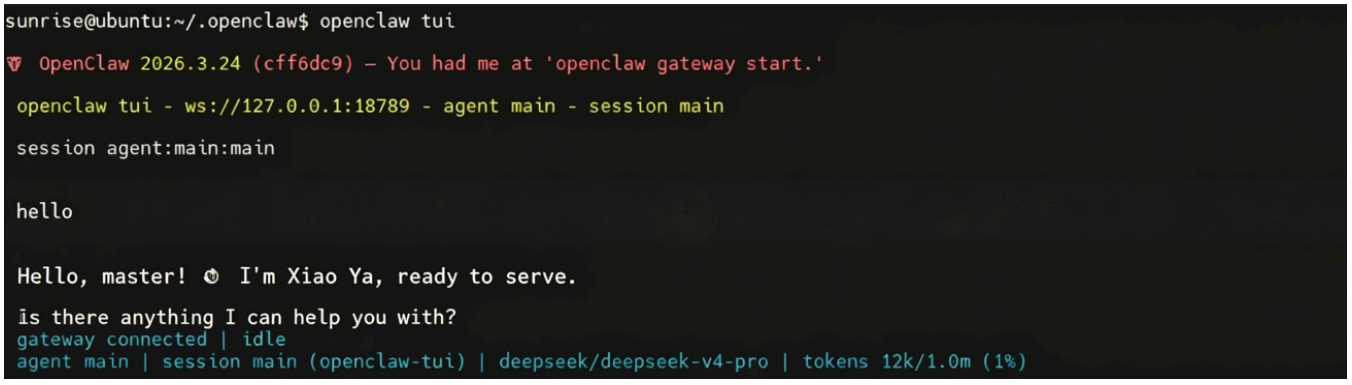

19:42
2
+86 153 3885 7526 (你)
给自己发消息
error-code#overdue-payment
19:40
Make the servo swing left and right .
19:41
The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.
19:42
Done! The servo completed a left-to-right swing:
• Moved to 0° (left)
• Swung to 180° (right)
• Returned to 90° (center)
The servo is now ready for the next command.
19:42
I
The
I'm
Q W E R T Y U I O P
A S D F G H J K L




## 5.2 TUI Method

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue chatting:

```
openclaw tui
```

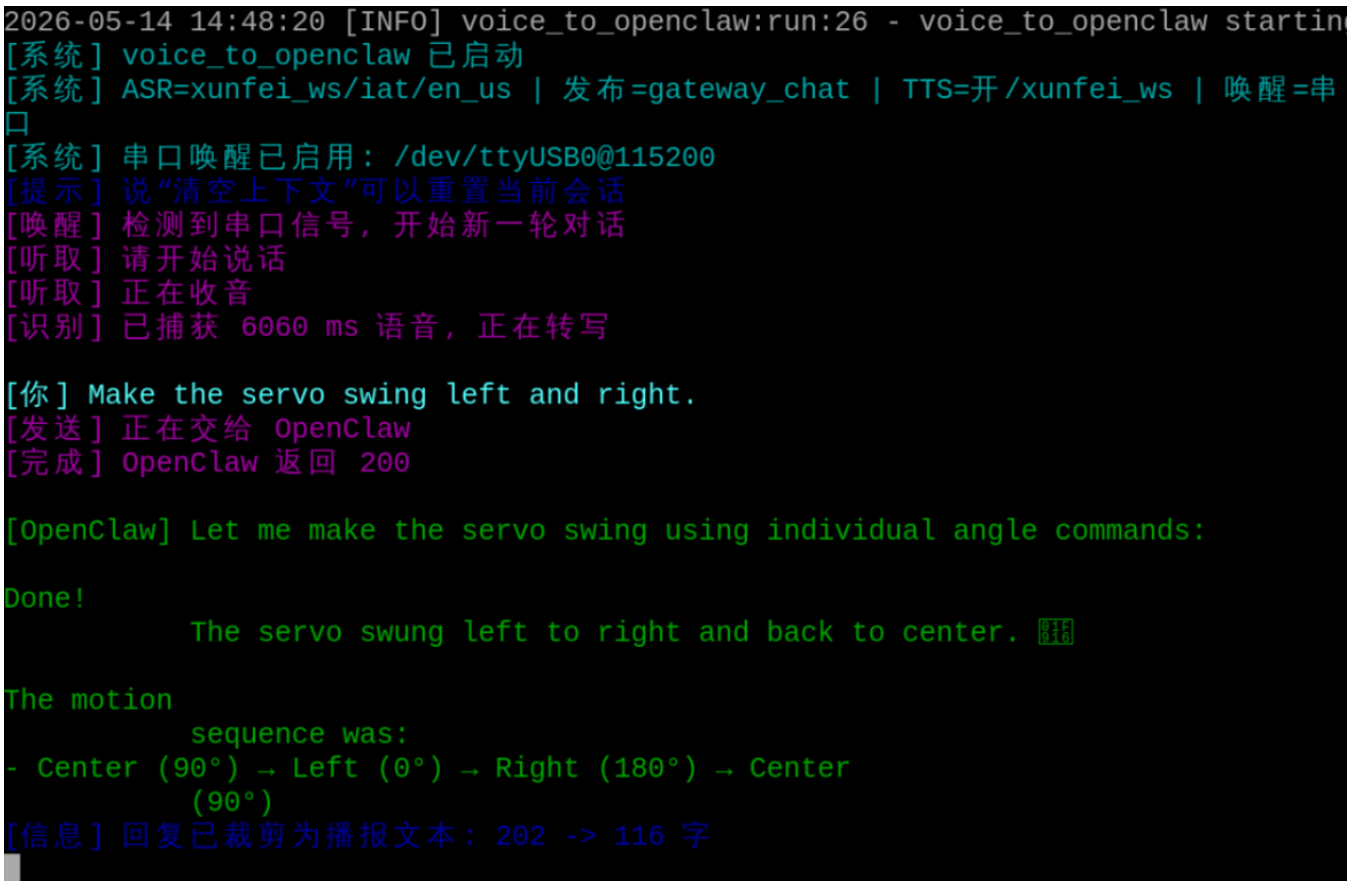



## 5.3 Voice Code Method

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake process, please refer directly to [05-AI Voice Interaction Configuration]. Voice mode is more suitable for demonstrating spoken servo commands, for example:

- Turn the servo to 90 degrees
- Make the servo nod
- Make the servo swing left and right




# 6. Comprehensive Cases

Note: The following are all demonstrated using `openclaw tui` for chat (you can choose your preferred chat method)

## 6.1 Case 1: Servo Dancing

### Experimental Objective

Make the servo move continuously according to a set of preset angles, creating a rhythmic "dancing" effect.

### Effect Description

The servo can switch between multiple angles, such as swinging left, centering, swinging right, rapid shaking, and then returning to the initial position. Even with just one servo, the changes in movement rhythm can still create a noticeable "dancing feel".

### Example Commands

- Make the servo dance for 10 seconds
- Make the servo swing left and right, with a faster rhythm
- Arrange a three-beat dance step for the servo

## 6.2 Case 2: Yes/No Q&A

### Experimental Objective

Make the servo express "yes" or "no" with movements, such as nodding for "yes" and shaking for "no".

### Effect Description

When OpenClaw determines the answer is affirmative, the servo performs a nodding action; when the answer is negative, the servo performs a head-shaking action. This case is very suitable for highlighting the combination of "natural language understanding + action expression".

### Example Commands

- If the answer is "yes", make the servo nod
- If the answer is "no", make the servo shake its head
- Help me judge this question and answer with servo movements

## 6.3 Case 3: Simulated Table Tennis

### Experimental Objective

Make the servo continuously move back and forth, simulating the feeling of a ping pong ball bouncing back and forth across the table.

### Effect Description

The servo oscillates at a constant or variable speed between the left and right boundary angles. If accompanied by voice descriptions or screen prompts, the audience will easily understand this as a "ping pong ball back and forth" simulation.

### Example Commands

- Make the servo simulate a ping pong ball bouncing back and forth
- Swing left and right 20 times, with a faster speed
- Start from the left, do a back and forth movement demo

## 6.4 Case 4: Simulated Pointer

### Experimental Objective

Use the servo as a rotatable pointer to indicate values, directions, or states.

### Effect Description

This case can very intuitively demonstrate the mapping from "abstract values -> actual angles". For example, mapping values from 0 to 100 to 0 to 180 degrees, the servo can point to the target position like a gauge needle.

## Example Commands

- Turn the pointer to 30%
- Make the servo point to the 90 degree position
- Simulate a dashboard, the current value is 75